2) $H(s) = \frac{1}{s^2 + 2\alpha + \omega_0^2}$, $u(t-a) * h(t) = x(t)$

$$H(s) = \frac{1}{(s+\alpha)^2 + \omega_0^2 - \alpha^2} = \frac{1}{(s+\alpha)^2 + \omega_d^2}$$

$$\mathcal{L}^{-1}\{H(s)\} = h(t) = \frac{1}{\omega_d} e^{-\alpha t} \sin(\omega_d t)\, u(t)$$

Steps thru prev. values to find value at t.

$$x(t) = \int_0^{\infty} u(t) * h(t-\tau)\, d\tau$$

$$= \int_0^{t} h(t-\tau)$$

$$= \frac{1}{\omega_d} \int_0^{t} e^{-\alpha(t-\tau)} \sin(\omega_d(t-\tau))\, dt$$

WOLFRAM: $x(t) = \frac{1}{\alpha^2 + \omega_d^2}\left[1 - e^{-\alpha t}\left(\cos(\omega_d t) + \frac{\alpha}{\omega_d}\sin(\omega_d t)\right)\right]$

$$x(t) = \begin{cases} 0, & t < a, \\ \frac{1}{\alpha^2 + \omega_d^2}\left[1 - e^{-\alpha(t-a)}\left(\cos(\omega_d(t-a)) + \frac{\alpha}{\omega_d}\sin(\omega_d(t-a))\right)\right] & t \geq a \end{cases}$$

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$$\cos(x) + \frac{\alpha}{w_d}\sin(x) = R\cos(x-\phi)$$

Where: $R = \sqrt{1 + \left(\frac{\alpha}{w_d}\right)^2}$, $\phi = \tan^{-1}\left(\frac{\alpha}{w_d}\right)$

$$x(t) = \begin{cases} 0 & t < a \\ \frac{1}{\alpha^2 + w_d^2}\left[1 - Re^{-\alpha(t-a)}\cos[w_d(t-a) - \phi]\right] u(t-a) & t \geq a \end{cases}$$

Numerical: $w_0 = 1$, $\alpha = 0.2\,w_0 = 0.2$, $\Delta t = 0.1$, $a = 5$

$$h(t) = \frac{1}{w_d} e^{-\alpha t}\sin(w_d t)\,u(t)$$

$$= \frac{1}{\sqrt{1-0.2^2}} e^{-0.2t}\sin\left(\sqrt{1-0.2^2}\,t\right)u(t)$$

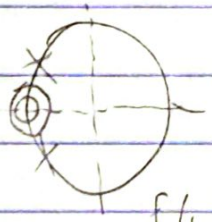
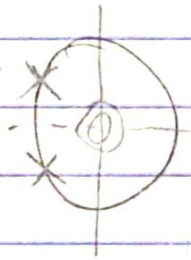
TO MATLAB ☺

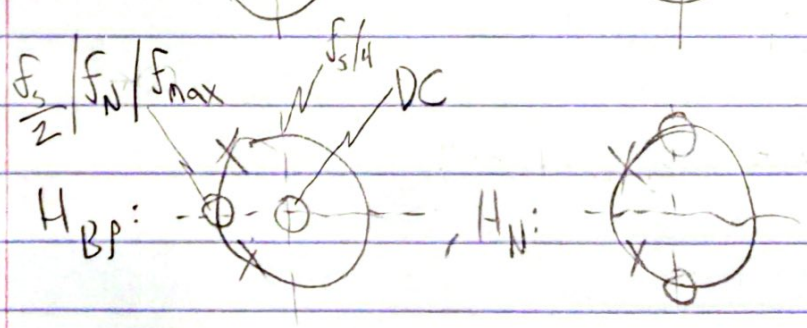
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3)

$$H(s) = \frac{N}{s^2 + 2\alpha s + \omega_o^2}$$

$$N = \begin{cases} H_{LP}: k\omega_o^2 \\ H_{HP}: ks^2 \\ H_{BP}: 2k\alpha s \\ H_N: k(s^2+\omega_o^2) \end{cases}$$

H_{LP}:  , H_{HP}: 



H_{LP}: Attenuates f_{max} with two zeroes

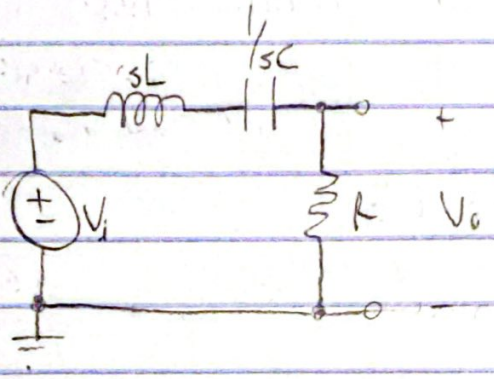
H_{HP}: Attenuates DC with two zeroes

H_{BP}: Attenuates DC & f_{max}

H_N: Attenuates mid-range.

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4)



$$\frac{V_o}{V_i} = \frac{R}{sL + \frac{1}{sC} + R} = \frac{sR/L}{s^2 + s\frac{R}{L} + \frac{1}{LC}} \qquad \alpha = \frac{R}{2L}$$

(2α ↑, ω_o^2 ↑)

$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-6}C}} = 10^6 = \frac{1}{\sqrt{10^{-12}}} \rightarrow C = 10^{-6}\ F$$

$C = 1\mu F$

$$|H(j\omega_o)| = \frac{j\omega_o \frac{R}{L}}{-\omega_o^2 + j\omega_o\frac{R}{L} - \omega_o^2}, \quad |H(j\omega_o)| = 1 \text{ for any } R$$

5) $\omega_o = 1$ rad/s, slope change

$\tau = 1/10$ s

$Q = 5$

$k = 1$

6.1) True, from HW5 I derived:

$$H(s) = \frac{1/(R_1R_2C_1C_2)}{s^2 + s(1/R_1C_1 + 1/R_2C_2) + 1/(R_1R_2C_1C_2)}$$

$$\frac{k}{(s+a)^2 + w_d^2} \rightarrow \mathcal{L}^{-1}\{H(s)\} = \frac{k}{w_d} e^{-at} \sin(w_d t)\, u(t)$$

6.2) False, L_1 & L_2 can be modeled as one larger inductor, only 2 resonant freq.

6.3) Yes, resonant freq. definition, shorts LC.

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$s \to 0$, $k = 1$, 2 poles

$$\frac{V_o}{V_i} = \frac{1/sC}{R + sL + 1/sC} = \frac{1/LC}{s^2 + \frac{R}{L} + \frac{1}{LC}}$$ LEFT

$$\frac{V_1 - V_i}{R_1} + (V_1 - V_o)sC_1 + \frac{V_1 - V_o}{R_2}$$

$$\frac{V_o - V_1}{R_2} + V_o sC_2 = 0 \Rightarrow V_o(1/R_2 + sC_2) = V_1/R_2$$

$$V_1 = V_o(sR_2C_2 + 1)$$

$$\frac{V_o(sR_2C_2 + 1) - V_i}{R_1} + (V_o(sR_2C_2 + 1) - V_o)sC_1 + \frac{(V_o(sR_2C_2 + 1) - V_o}{R_2}$$

$$\frac{V_o(sR_2C_2 + 1) - V_i}{R_1} + s^2 V_o R_2 C_1 C_2 + sV_o C_2$$

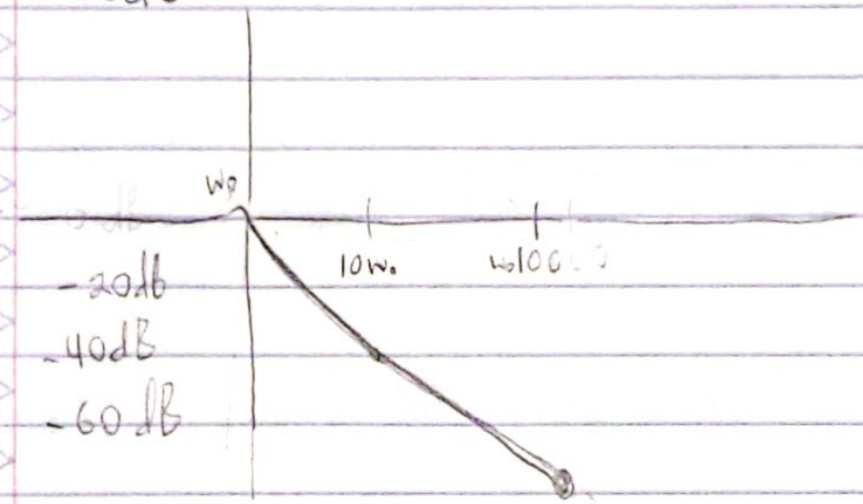
$$V_o\left[s^2 R_1 R_2 C_1 C_2 + s(R_1 C_2 + R_2 C_2 + 1)\right] = V_i$$

$$\frac{V_o}{V_i} = \frac{1/R_1 R_2 C_1 C_2}{s^2 + s\frac{R_1 + R_2}{R_1 R_2 C_1} + \frac{1}{R_1 R_2 C_1 C_2}} \Rightarrow s \to 0,\ k = 1,\ 2 \text{ poles}$$

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-40.1B/dec

Bode:





$$\frac{V_0 - V_a}{R_2} + V_0 s C_2 = 0$$

$$V_0\left(\frac{1}{R_2} + sC_2\right) = \frac{V_a}{R_2} \rightarrow V_a = V_0(sR_2C_2 + 1)$$

$$(V_a - V_i)sC_1 + \frac{V_a - V_0}{R_2} + \frac{V_a}{R_1} = 0$$

$$V_a\left(sC_1 + \frac{1}{R_2} + \frac{1}{R_1}\right) - \frac{V_0}{R_2} = V_i sC_1$$

$$V_0(sR_2C_2 + 1)\left(sC_1 + 1/R_2 + 1/R_1\right) - V_0/R_2 = V_i sC_1$$

$$= V_0\left(s^2R_2C_1C_2 + sC_2 + s\frac{R_2}{R_1}C_2 + sC_1 + 1/R_2 + 1/R_1 + 1/R_2\right)/sC_1 = V_i$$

$$= V_0\left(sR_2C_2 + \frac{C_2}{C_1} + \frac{R_2C_2}{R_1C_1} + 1 + 1/sR_1C_1\right) = V_i$$

$$H(s) = \frac{1}{sR_2C_2 + \frac{C_2}{C_1} + \frac{R_2C_2}{R_1C_1} + 1 + 1/sR_1C_1} = \frac{s/R_2C_2}{s^2 + s/R_2C_1 + s(1/R_1C_1) + s(1/R_2C_2) + 1/R_1R_2C_1C_2}$$

$$= \frac{s/R_2C_2}{s^2 + s(1/R_1C_1 + 1/R_2C_1 + 1/R_2C_2) + 1/R_1R_2C_1C_2}$$

$R_1 = R_2 = 1K$

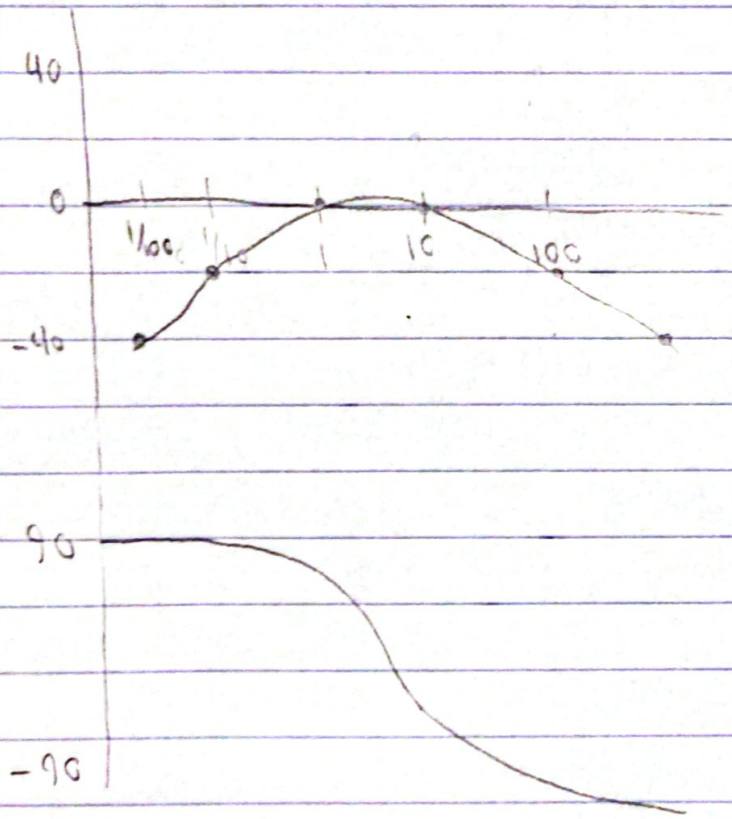
$C_1 = C_2 = 1mF$

$$= \frac{s}{s^2 + s(1+1+1) + 1} = \frac{s}{s^2 + 3s + 1}$$

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Bode

$W_c = 1$



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